INCH-POUND

MIL-H-5606F
AMENDMENT 1
26 August 1991

MILITARY SPECIFICATION

HYDRAULIC FLUID, PETROLEUM BASE; AIRCRAFT, MISSILE AND ORDNANCE

This amendment forms a part of MIL-H-5606F, dated 6 December 1990, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 3

Line 9, add:

"ASTM D 4636 Standard Test Method for Corrosiveness and Oxidation Stability of Hydraulic Oils, Aircraft Turbine Engine Lubricants, and Other Highly Refined Oil"

PAGE 8

Para 4.3.3.1, delete and substitute:

"4.3.3.1 Toxicity coordination. At the time of qualification submittal, the contractor shall provide the following certified materials for each ingredient to the Air Force Occupational and Environmental Health Directorate, AL/OEMB, Brooks AFB, TX 78235:

Chemical Name and Formula (nomenclature of the International Union of Pure and Applied Chemistry)

NIOSH Identification Number (Accession or identification number referenced in the Registry of Toxic Effects of Chemical Substances, if assigned)

In addition, the range of percentages of harmful or toxic components in the finished product and any toxicological data useful in evaluating the safety of the formulation shall be furnished. The Government reserves the right to determine whether such data is adequate for the purposes of qualification under the provisions of this specification. The AL/OEMB personnel have 15 working days from the receipt of the contractor materials to report compliance/noncompliance to the qualifying agency".

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Para 4.4, delete and substitute:

"4.4 Quality conformance inspection. The quality conformance inspection shall consist of sampling plans A, B, C and D. Quality conformance inspection shall consist of examination of the sample of filled containers (see 4.5.8), determination of the solid particle contamination count (see 4.5.6), and testing the sample against all requirements specified in section 3, except for corrosiveness and stability (oxidation, shear, and storage) and swelling of synthetic rubber. In the event of a solid particle contamination count failure, the referee method shall be the automatic particle counter method calibrated with latex spheres. Samples shall be labeled completely with information identifying the purposes of the sample, name of product, specification number, lot and batch number, date of sampling and contract number."

PAGE 11

Para 4.5.5 b, 2nd line, after 30 minutes add: "0°C."

PAGE 12

Table VII, delete and substitute:

"TABLE VII. Test methods for hydraulic fluid properties.

Characteristic	Test Method FED-STD-791	ASTM
Specific gravity		D 287
Viscosity		D 445
Pour point		D 97
Flash point		D 93
Acid or base No.		D 664
Color		D 1500
Corrosiveness and oxidation stability		D 4636 1/
Low temperature stability	3459	
Swelling of synthetic rubber	3603	
Evaporation		D 972 2/
Solid particle size	3009 3/	
Filtration time	3009 4/	
Foaming characteristics		D 892
Water		D 1744
Storage stability	3465	
Steel-on-steel wear		D 4172 5/
Chlorine concentration 6/		

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1/ Test shall be run for 168 hours at 135°C.

2/ Test shall be run for 6 hours at 71°C.

3/ Particle contamination shall be measured by the use of automatic particle counters. Directions in the manual for the respective instrument shall be followed.

4/ Filtration time shall be measured using a single membrane filter.

5/ Condition B.

6/ Determined by using MIL-STD-1844 procedure."

PAGE 13

Para 4.5.6, lines 2 and 3, delete "ASTM E 313" and substitute "ASTM F 313".

PAGE 16

Para 6.3, line 8, delete "WRDC/MLSE" and substitute "WL/MLSE".

Para 6.4.1, 6.4.2, delete and substitute:

"6.4.1 The sample (1 pint) of shear stability reference fluid for the test specified in 4.5.5 may be obtained from WL/MLSE, Wright-Patterson AFB OH 45433-6533.

6.4.2 Samples of standard synthetic rubber NBR-L shall subscribe to the formulation in accordance with AMS 3217/2. A suggested source (material) is Wynn's Precision Rubber Products Corporation, Hartman Drive, Lebanon TN 37087."

Custodians:
Army - ME
Navy - AS
Air Force - 11

Review activities:
Army - MI, SM, AR
Navy - SA, SH, OS
Air Force - 68
DLA - GS

Preparing activity:
Air Force - 11

Proj No. 9150-1095

International interest:
(See section 6)

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